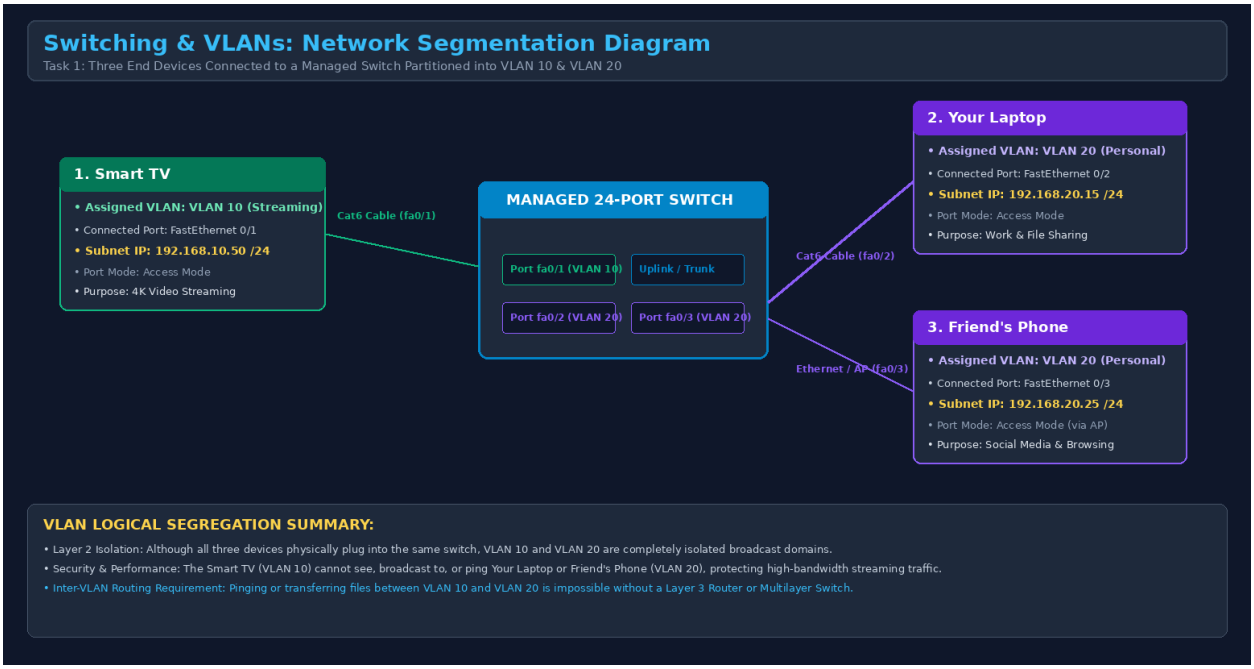
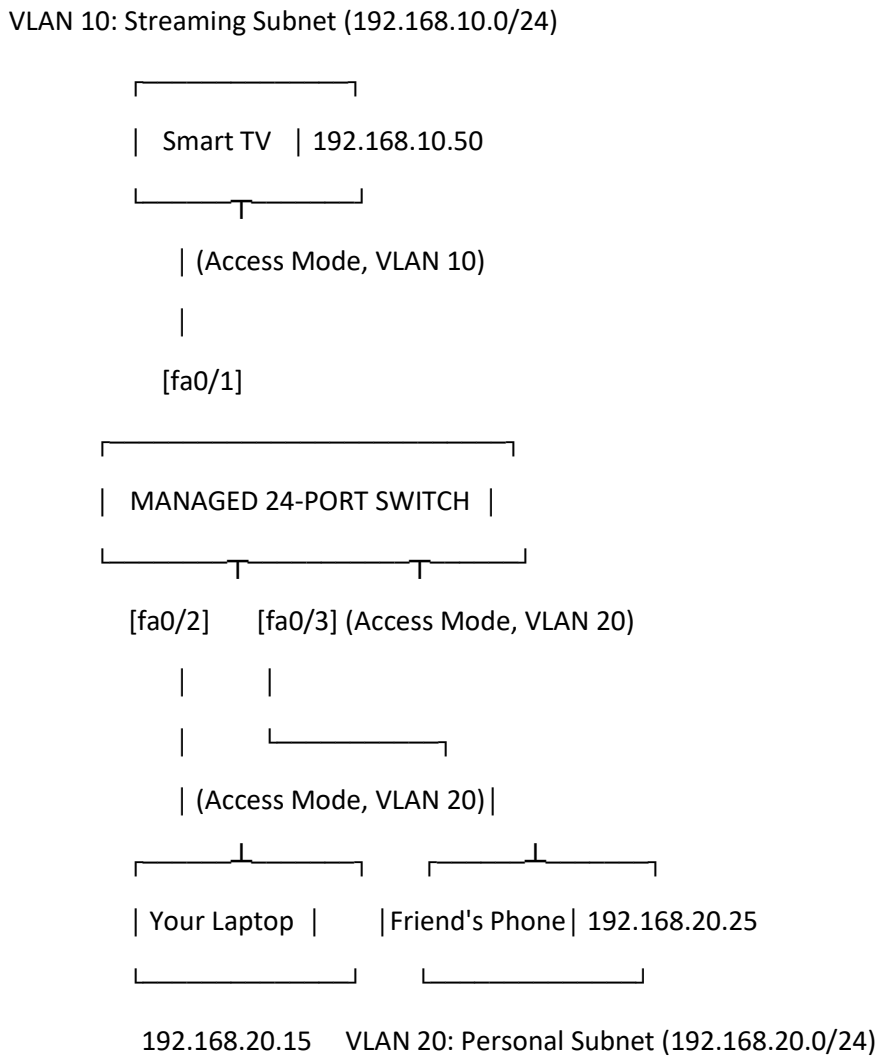
# Switching and VLANS

**Task 1: Switch VLAN Network Diagram**

A diagram illustrating the logical VLAN isolation over a shared 24-port physical switch:






**Port to VLAN Assignments:**

- **VLAN 10 (Streaming Network):** Dedicated Port **fa0/1** →\rightarrow→ Connected to the **Smart TV** (192.168.10.50).
- **VLAN 20 (Personal Subnet):** Ports **fa0/2** and **fa0/3** →\rightarrow→ Connected to **Your Laptop** (192.168.20.15) and **Friend's Phone** (192.168.20.25).

**Operational Feature:** Because they are segregated into two distinct Layer 2 broadcast domains, the laptop cannot communicate with or ping the TV even though they are plugged into the same switch.

**Task 2: Configure VLAN 10 & VLAN 20 on a Simulated Switch**

To execute this configuration in **Cisco Packet Tracer**, connect five PCs to a switch and use the Command Line Interface (CLI) of the switch (Switch>):

Code snippet

```
Switch> enable
Switch# configure terminal

! Create VLAN 10 (Streaming) and name it
Switch(config)# vlan 10
Switch(config-vlan)# name Streaming
Switch(config-vlan)# exit

! Create VLAN 20 (Personal Devices) and name it
Switch(config)# vlan 20
Switch(config-vlan)# name Personal
Switch(config-vlan)# exit

! Assign FastEthernet 0/1, 0/2, and 0/3 to VLAN 10 (3 PCs)
Switch(config)# interface range fa0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

! Assign FastEthernet 0/4 and 0/5 to VLAN 20 (2 PCs)
Switch(config)# interface range fa0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end

! Save the config
```

Switch# write memory

**To Verify Configuration:**

Run the show vlan brief command:

Code snippet

Switch# show vlan brief

VLAN Name Status Ports

---- -------------------------------- --------- -------------------------------

1 default active Fa0/6, Fa0/7, Fa0/8, Fa0/9...

10 Streaming active Fa0/1, Fa0/2, Fa0/3

20 Personal active Fa0/4, Fa0/5

- **VLAN Separation Verification Test:**

If you try to ping from PC 1 (192.168.10.10 on VLAN 10) to PC 4 (192.168.20.40 on VLAN 20), the command prompt will show:

**Request timed out.** (Ping fails because they reside on separate VLANs without a router).

**Task 3: Move a PC from VLAN 10 to VLAN 20**

To relocate the third host connected to port FastEthernet 0/3 from the Streaming network over to the Personal network, execute the following commands in Cisco Packet Tracer:

**1. "Before" Configuration Check:**

Verify the interface status and current VLAN assignment:

Code snippet

Switch# show interfaces fa0/3 switchport

Name: Fa0/3

Switchport: Enabled

Administrative Mode: static access

Access Mode VLAN: 10 (Streaming)

**2. Re-Assignment Commands:**

Code snippet

Switch> enable

Switch# configure terminal

Switch(config)# interface fa0/3

Switch(config-if)# switchport access vlan 20

Switch(config-if)# end

Switch# write memory

**3. "After" Configuration Check:**

Verify the new VLAN registration:

Code snippet

Switch# show interfaces fa0/3 switchport

Name: Fa0/3

Switchport: Enabled

Administrative Mode: static access

Access Mode VLAN: 20 (Personal)

**4. Connectivity Test Outcome:**

- **Before the Move:** This PC was in VLAN 10, meaning it could ping PC 1 and PC 2 (VLAN 10), but could not ping PC 4 and PC 5 (VLAN 20).

- **After the Move:** This PC's IP address must be updated to the VLAN 20 range (e.g., changing from 192.168.10.30 to 192.168.20.30). After the update, it can successfully ping PC 4 and PC 5, but pinging VLAN 10 now fails.

**Task 4: Automation Script for VLAN 30 on 10 Ports**

To avoid retyping the port assignment commands ten times, use Cisco's interface range command:

Code snippet

```
Switch> enable
Switch# configure terminal

! Create the destination VLAN first
Switch(config)# vlan 30
Switch(config-vlan)# name Guest_Workstations
Switch(config-vlan)# exit

! Automate ports assignment using interface range
Switch(config)# interface range fa0/10 - 19
Switch(config-if-range)# description Guest VLAN Ports
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# no shutdown
Switch(config-if-range)# end

! Save Changes
Switch# write memory
```

**Command Explanations:**

- **interface range fa0/10 - 19:** Target and execute commands across 10 interfaces simultaneously (specifically ports fa0/10, fa0/11... up to fa0/19).
- **switchport mode access:** Configures the interfaces to connect directly to end-user devices (clients) and prevent trunking negotiations.
- **switchport access vlan 30:** Maps all 10 ports directly to VLAN 30.
- **no shutdown:** Ensures the ports are administratively active. Use show vlan brief to verify that VLAN 30 is created and ports Fa0/10 through Fa0/19 are assigned.